Лекция 6.1 PDDL

PDDL

- Planning Domain Definition Language
- PDDL 1.2, PDDL 2.1, PDDL 2.2, PDDL 3.0, PDDL 3.1
- Standard encoding language for “classical” planning tasks

Основные сущности языка

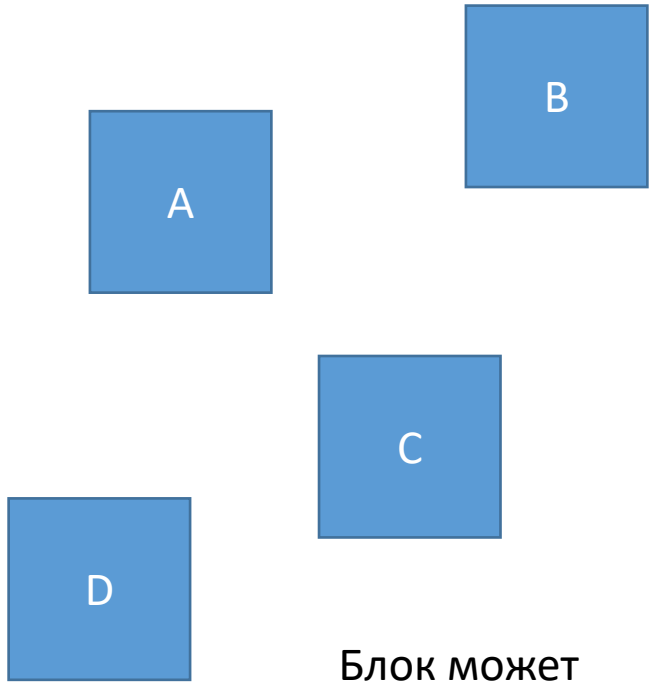
- Описание проблемы(problem file):
  - Объекты – Objects
  - Начальное состояние – Initial state
  - Целевое состояние – Goal specification
- Описание предметной области (domain file):
  - Предикаты – Predicates
  - Операторы перехода – Actions/Operators
- Дополнительно:
  - Требования – Requirements – аналог include
  - Типы – Types

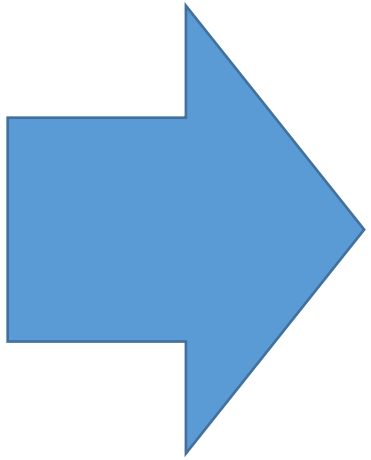
Синтаксис

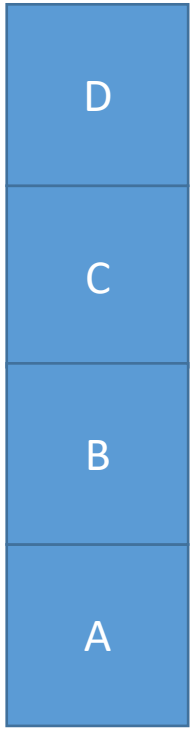
```
(define (domain <domain name>)
   <PDDL code for requirements>
   <PDDL code for types>
   <PDDL code for predicates>
   <PDDL code for first action>
   [...]
   <PDDL code for last action>
)
```

```
(define (problem <problem name>)
  (:domain <domain name>)
  <PDDL code for objects>
  <PDDL code for initial state>
  <PDDL code for goal specification>
)
```

Пример задачи – разложить кубики друг на друга







Блок может

- Лежать на столе
- Лежать на другом блоке
- Быть в руке у человека
- Быть свободным сверху
- Быть занятым сверху

Рука может

- Быть свободной
- Быть занятой

Пример задачи – разложить кубики друг на друга

```
(define (domain BLOCKS)
  (:requirements :strips :typing)
  (:types block)
  (:predicates (on ?x - block ?y - block)
(ontable ?x - block) (clear ?x - block)
(handempty)  (holding ?x - block))

(:action pick-up
              :parameters (?x - block)
              :precondition (and (clear ?x)
(ontable ?x) (handempty))
              :effect
              (and (not (ontable ?x))
                            (not (clear ?x))
                            (not (handempty))
                            (holding ?x)))
(:action put-down
              :parameters (?x - block)
              :precondition (holding ?x)
              :effect
              (and (not (holding ?x))
                            (clear ?x)
                            (handempty)
                            (ontable ?x)))
```

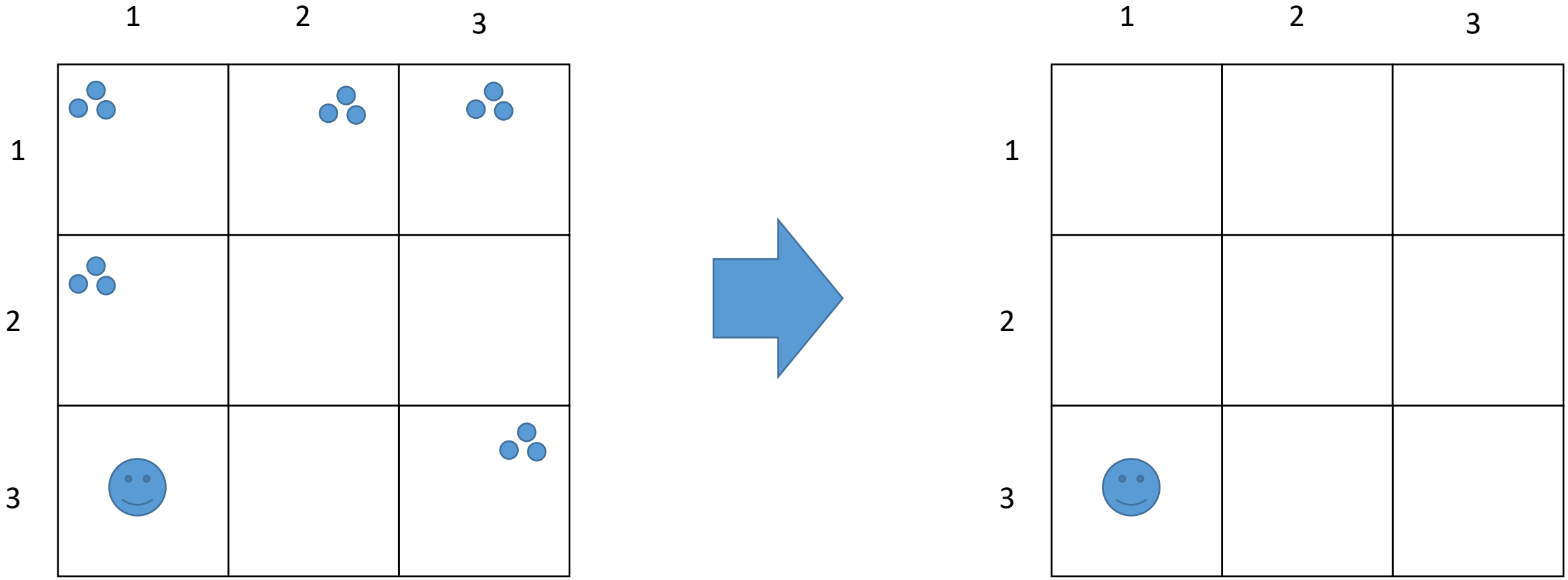
```
(:action stack
              :parameters (?x - block ?y - block)
              :precondition (and (holding ?x)
(clear ?y))
              :effect
              (and (not (holding ?x))
                            (not (clear ?y))
                            (clear ?x)
                            (handempty)
                            (on ?x ?y)))
 (:action unstack
              :parameters (?x - block ?y - block)
              :precondition (and (on ?x ?y)
(clear ?x) (handempty))
              :effect
              (and (holding ?x)
                            (clear ?y)
                            (not (clear ?x))
                            (not (handempty))
                            (not (on ?x ?y)))))
```

```
(define (problem BLOCKS-4-0)
(:domain BLOCKS)
(:objects D B A C - block)
(:INIT (CLEAR C) (CLEAR A) (CLEAR B)
(CLEAR D) (ONTABLE C) (ONTABLE A)
 (ONTABLE B) (ONTABLE D)
(HANDEMPTY))
(:goal (AND (ON D C) (ON C B) (ON B
A)))
)
```

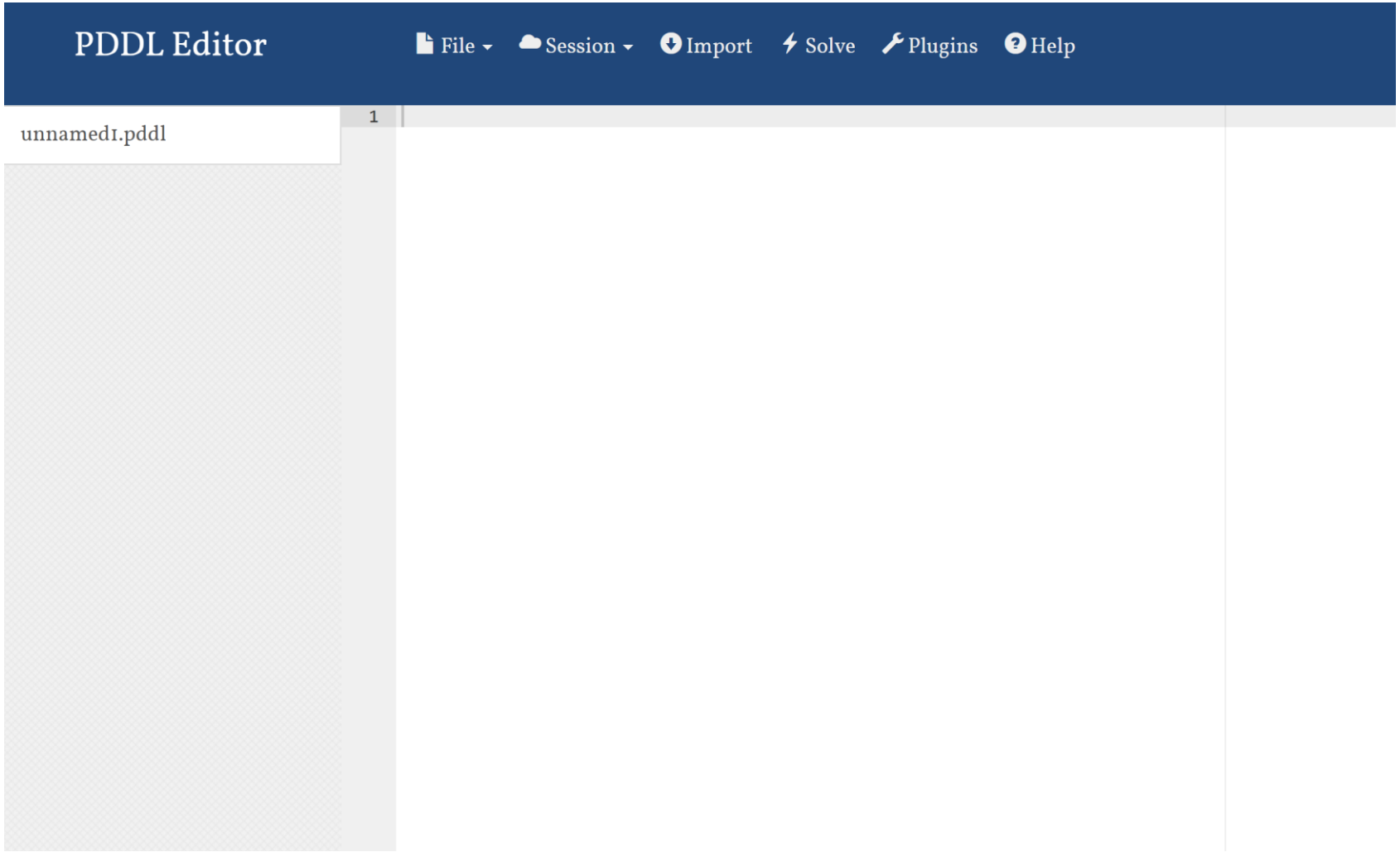
```
(pick-up b)
(stack b a)
(pick-up c)
(stack c b)
(pick-up d)
(stack d c)
```

http://www.cs.toronto.edu/~sheila/2542/w09/A1/introtopddl2.pdf
https://github.com/pellierd/pddl4j/wiki/A-tutorial-to-start-with-PDDL

Л.р. 4 планирование уборки роботом



Вариант, где запускать



http://editor.planning.domains/